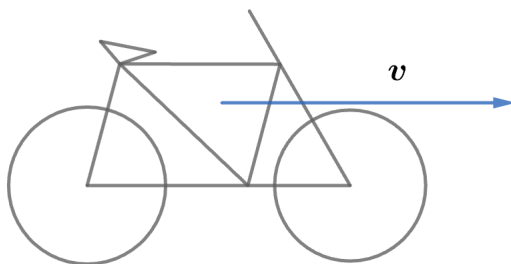


2011 F=ma Exam: Problem 1

Kevin S. Huang



Let t be the time the cyclist traveled at 22 km/hr. Then by definition of average speed,

$$d = (22 \text{ km/hr})t = (17.5 \text{ km/hr}) \left(t + \frac{1}{3} \text{ hr} \right)$$

Solving for the time,

$$4.5t = 17.5/3$$

$$t = 1.296 \text{ hr}$$

Thus,

$$d = (22 \text{ km/hr})(1.296 \text{ hr}) = 28.5 \text{ km}$$

so the answer is A.